# Introduction to Transaction Control

Oracle server ensures data consistency based upon transactions. Transactions consist of DML statements that make up one consistent change tothe data.

**Transaction Start and End Scenarios:**

A Transaction begins when the first executable sql statement is encountered. The Transaction terminates when the following specifications occur.

- A COMMIT or ROLL BACK issued.
- A DDL statement issued
- A DCL statement issued
- Failure of machine or system crashes
- A DDL or DCL statement is automatically committed and hence implicitly ends a Transaction.

**Explicit Transaction Control Statements**

- COMMIT
- SAVEPOINT
- ROLLBACK
- COMMIT

It ends the current Transaction by making all pending data changes permanent.

**Syntax:**COMMIT;

Once commit is issued, data changes will become permanent.
The previous state of the data is permanently lost.
All users can view the results of the recent transactional changes.

**Ex**: UPDATE employee SET salary =1000 WHERE emp_id = 10;
COMMIT;

**SAVEPOINT:**

It marks a savepoint with in the current Transaction. We can create multiple savepoints in single Transaction.Savepoints can be used to control the reverting of changes.

**Syntax**:SAVEPOINT ;
Ex: SAVEPOINT S1;

**ROLLBACK:**

- It ends the current Transaction by discarding all pending data changes.
- The data changes are undone.
- The previous state of data is restored.
- The locks on the affected rows are automatically released.

**Syntax:** ROLLBACK or ROLLBACK to ;

**Ex**: UPDATE employee SET salary =1000 WHERE emp_id = 10;
ROLLBACK;

Using savepoint the Transaction can be discarded up to the marker by using rollback statement.

**Ex**: INSERT INTO employee VALUES(10,'JHON',3000);
INSERT INTO employee VALUES(10,'KELLY',2000);
SAVEPOINT S1;
INSERT INTO employee VALUES(10,'WILSON',4000);
ROLLBACK TO S1;